

CFD MODELING OF OPERATING PROCESSES OF A SOLAR AIR HEATER IN ANSYS FLUENT

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Rapid development of the technologies of use of renewable energy sources in the recent decade has opened up prospects for satisfying our society's ever growing needs for fuel and power resources. In the present work, the author has shown results of investigation into the thermal and aerodynamic characteristics of a solar air heater. A computer model whose adequacy has been confirmed using experimental data was developed in the ANSYS Fluent software product. It has been determined that for the end of calendar autumn, the operating efficiency of the solar air heater is the highest at the angle of inclination of the absorption surface to the earth $\alpha = 60^\circ$. The dependence of the value of heating of air in the solar heater during daylight hours has been determined in the physical and computer experiment; it has been established that for the city of Samara (GMT + 4) the maximum air heating is observed in the interval between 13.00 and 14.00. The influence of transverse ribs on the heat exchange between the light-absorbing surface and the air has been shown. A curvilinear shape of the temperature contour is formed in the immediate vicinity of the ribs due to the formation of vortices and reverse-current zones.

Keywords: ANSYS, CFD modeling, renewable energy sources, solar air heater, energy efficiency.

Introduction. Solar radiation is a promising energy source capable of satisfying our society's ever growing needs for fuel and power resources without any substantial adverse environmental impacts [1]. Solar energy has long been used in various fields of the national economy: heat and electric power supply systems [2], agriculture and healthcare, the energetics of heat technologies, transport [3], and others.

In recent decades, there has been a substantial qualitative and quantitative development of the technologies of use of solar energy for electric power supply needs. Solar air heaters (SAHs) have begun to find wide application in heat supply systems owing to their simple structure and low capital and operating costs. The surface absorbing solar radiation converts the solar radiation into the useful thermal energy of air: the SAH basic heat-transfer agent.

A regular smooth absorbing plate has a relatively low thermal efficiency because of the low coefficient of heat transfer to the flow air. The efficiency of heat transfer in the SAH can be improved by implementation of the following measures: (1) increase in the share of the useful area of the absorption surface and (2) enhancement of the air-flow turbulence and formation of vortices/reverse-current zones. These measures are implemented due to the branched heat-transfer surface obtained by creating an obstacle in the region of the air flow. However, extra surfaces lead to an increase in the aerodynamic drag and hence to a greater spent work of the fan.

In the present study, we give computational fluid dynamics (CFD) modeling of the operating processes of an SAH in the ANSYS Fluent software product using the built-in module "Solar Calculator." On the basis of the developed model, thermal and aerodynamic characteristics of the SAH with transverse ribs have been investigated numerically. The adequacy of the developed model is confirmed by the results of physical experiment. In the work, we present a detailed description of the geometry, the computational mesh, the boundary conditions, settings of the Fluent solver, and also of the processing of obtained results and their validation with experimental data.

Characteristics of the Solar Air Heater. Geometry. The solar air heater incorporates the absorption surface (3), the inlet (2) and outlet (1) of the heat-transfer agent, and also the transparent surface manufactured from glass of thickness 3 mm. The SAH casing is fabricated from wood; the absorption surface with dimensions 1000×850 mm is covered with heat-insulating material on the outside. On the absorption surface, there are transverse ribs (5) manufactured from 25×25 mm bars of length 600 mm. Lateral walls (4) are fabricated from 50×50 mm bars. The diameters of the inlet and outlet air

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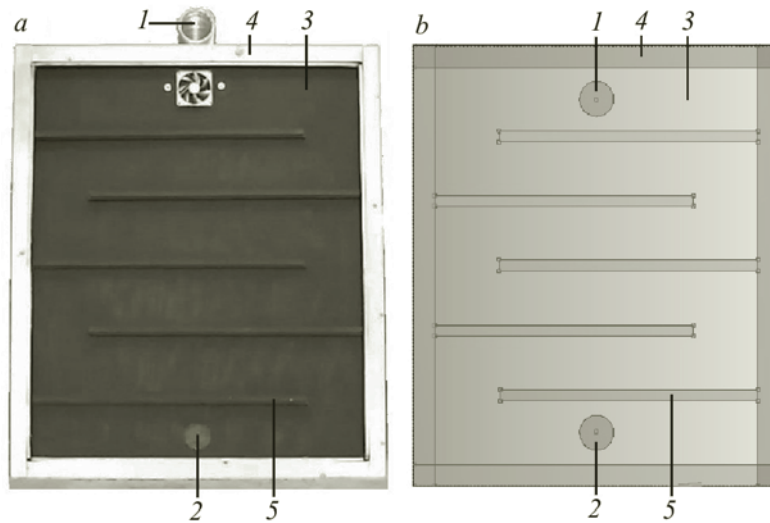
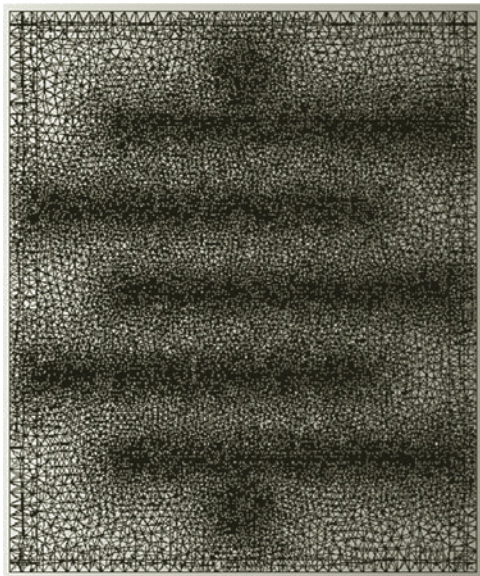


Fig. 1. Experimental setup (a) and the ANSYS Fluent calculated geometry (b) of an SAH: 1) air outlet; 2) air inlet; 3) light-absorbing surface; 4) lateral wall; 5) ribs.



Defaults		Inflation	
Physics Preference	CFD	Use Automatic Inflation	None
Solver Preference	Fluent	Inflation Option	Smooth Transition
Relevance	100	Transition Ratio	0,272
Sizing		Maximum Layers	5
Use Advanced Size Function	On: Curvature	Growth Rate	1,2
Relevance Center	Medium	Inflation Algorithm	Pre
Initial Size Seed	Active Assembly	View Advanced Options	No
Smoothering	Medium	Advanced	
Transition	Slow	Shape Checking	CFD
Span Angle Center	Fine	Element Midside Nodes	Dropped
Curvature Normal Angle	Default (12,0 °)	Straight Sided Elements	
Min Size	Default (2,0761e-002 cm)	Number of Retries	0
Max Face Size	Default (2,07610 cm)	Extra Retries For Assembly	Yes
Max Size	Default (4,15210 cm)	Rigid Body Behavior	Dimensionally Reduced
Growth Rate	Default (1,10)	Mesh Morphing	Disabled
Minimum Edge Length	2,50 cm	Defeaturing	
Inflation		Pinch Tolerance	Default (1,8685e-002 cm)
Use Automatic Inflation	None	Generate Pinch on Refresh	No
Inflation Option	Smooth Transition	Defeaturing Tolerance	Default (1,038e-002 cm)
Transition Ratio	0,272	Statistics	
Maximum Layers	5	Nodes	132329
Growth Rate	1,2	Elements	686806
Inflation Algorithm	Pre	Mesh Metric	None

Fig. 2. Computational mesh and its characteristics.

pipes are 80 mm. The calculated geometry of the computational model (Fig. 1b) fully follows the experimental setup (Fig. 1a).

Mesh. Construction of a computational mesh is performed in the built-in Meshing module. To upgrade the solution, the computational mesh is structured near the walls of the absorption surface. A 5-mm-thick wall boundary layer is created using the standard functions "Sizing" and "Inflation." The structure of the computational mesh and its characteristics are given in Fig. 2. The total number of mesh elements is 686,806.

Mathematical description. A mathematical description of the model in generated form is a system of equations that consists of the equations of continuity, of the law of conservation of momentum, and of the law of conservation of energy. To model the dynamics of air motion, use is made of Navier–Stokes equations closed by the standard k – ϵ turbulence model [4, 5]:

the continuity equation

$$\frac{\partial}{\partial x_i} (\rho u_i) = 0, \quad (1)$$

the equation of the law of conservation of momentum

$$\frac{\partial}{\partial x_i} (\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] + \frac{\partial}{\partial x_j} (-\rho u_i u_j), \quad (2)$$

and the heat-balance equation

$$\frac{\partial}{\partial x_i} (\rho u_j T) = \frac{\partial}{\partial x_j} \left[(a + a_t) \frac{\partial T}{\partial x_j} \right]. \quad (3)$$

Thermophysical properties of the air as well as of the wood and the glass have been obtained from the Fluent Database.

Initial and boundary conditions for the temperature are of the following form:

$$T_{in}(t = 0) = T_s(t = 0) = T_{gl}(t = 0) = T_{env}(t = 0), \quad (4)$$

$$\left. \frac{\partial T_{in}}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial T_{gl}}{\partial x} \right|_{x=H} = 0, \quad \left. \frac{\partial T_s}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial T_s}{\partial x} \right|_{x=H} = 0, \quad (5)$$

where T_{in} , T_s , T_{gl} , and T_{env} are the temperatures of the air at the heater inlet, of the light-absorbing surface, of the glass, and of the environment respectively, K.

The value of the solar radiation is calculated using the "Solar Calculator" device built into ANSYS, which permits determining the solar-radiation intensity as a function of the following parameters [6]: the geographical coordinates, the date and time of computer experiment, the angle of inclination of the SAH to the earth's surface, and the Sunshine Factor.

Results and Discussion. In the present study, the physical and computer experiments were conducted in the city of Samara on the 22nd of November, 2016, in the time interval from 10 to 16 h; the Sunshine Factor was equal to 1 (absolutely clear sky). The indicated conditions were entered in the solar calculator (Fig. 3).

The temperature contour of the light-absorbing surface (contour 1) and of the air at a distance of 14 mm from the surface of the glass (contour 2) for the conditions given in Fig. 3 is shown in Fig. 4. In Fig. 4, one can see a nonuniform character of the contour due to the presence of transverse ribs on the light-absorbing surface. The contour's curvilinear shape is formed in the immediate vicinity of the ribs due to the formation of vortices and reverse-current zones. The low-temperature region on the upper edge of the ribs points to the more intense heat exchange between the light-absorbing surface and the air, which is caused by the high turbulence of the flow in this zone. At the same time, the region of the surface between the ribs has a comparatively higher temperature as a consequence of the less intense heat exchange.

Temperature maxima are observed near the lateral walls which may be called a "dead zone" because of the insignificant motion of the heated air in this SAH part, which impairs substantially its operating efficiency [7].

The operating efficiency of the SAH depends substantially on the velocity of sound at entry, and also on the angle of its inclination to the earth's surface [8–10]. Figure 5 gives the plot of the difference of the air temperatures at the solar-heater inlet and outlet versus the inlet air velocity for different angles of inclination. To analyze the influence of these parameters on the value of air heating, we used, as the initial data, the following values: date and time — 22.1.2016, 13.00 (GMT + 4, the city of Samara); absolutely clear sky (Sunshine Factor = 1). According to the obtained data, the SAH operating efficiency is the highest for the angle of inclination of the absorption surface to the earth $\alpha = 60^\circ$. This is attributable to the fact that for the last part of November in the city of Samara, the sun is fairly low above the horizon and the cosine of the angle between the sunrays and the SAH surface has a minimum value. Clearly, with increase in the air velocity at entry into the SAH, the temperature of heating of the air decreases, since in the case the volumetric rate of flow of the heat-transfer agent increases. This fact is also confirmed by the model-experiment results given in Fig. 5.

Figure 6 shows the difference in air temperatures at the heater inlet and outlet during daylight hours for various inlet air velocities. The angle of inclination of the SAH to the earth's surface is $\alpha = 60^\circ$. The results of the computer experiment are shown by the lines, and the results of the physical experiment, by the symbols. Figure 6 demonstrates good agreement between the results of the computer and physical experiments. On the whole, we can speak of the adequacy of the developed

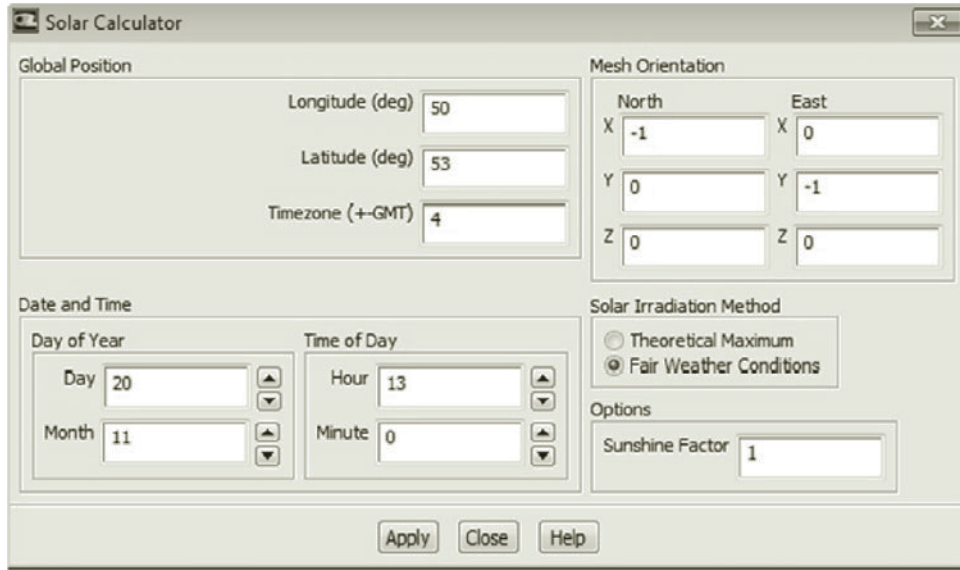


Fig. 3. Desktop of the solar calculator.

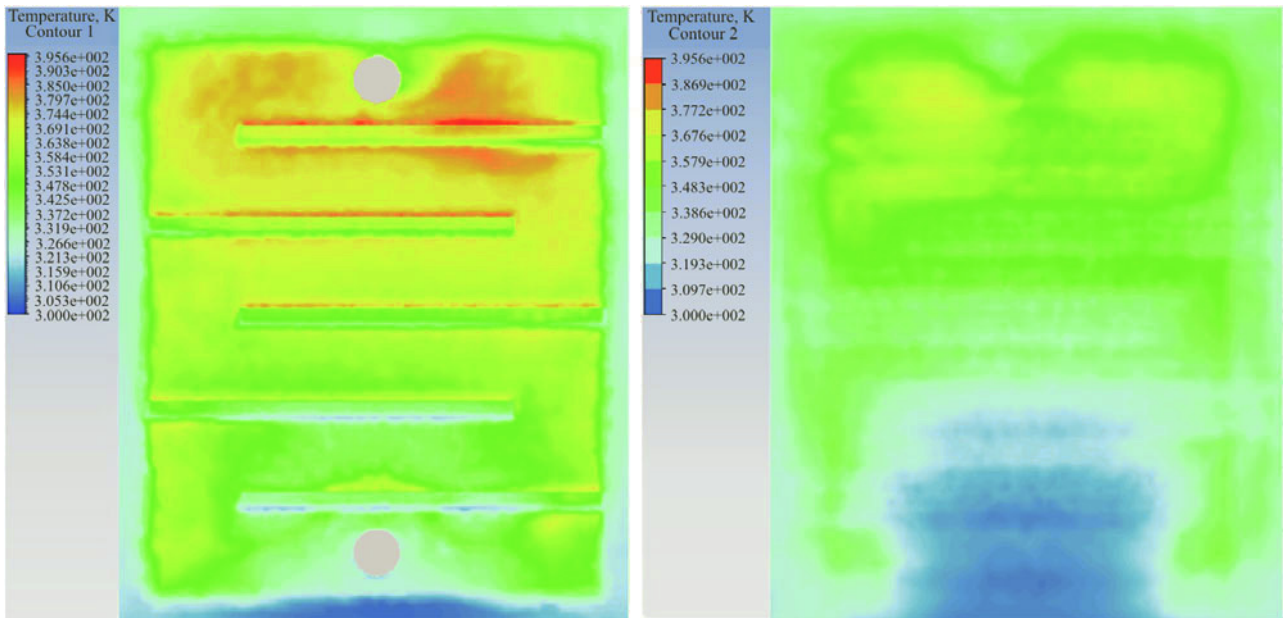


Fig. 4. Temperature contours of the light-absorbing surface (contour 1) and the air at a distance of 14 mm from the surface of the glass (contour 2) at $u_{in} = 1.5$ m/s for the conditions shown in Fig. 3.

computer model of an SAH to the actual SAH: the model's behavior is coincident with a sufficient degree of accuracy with the behavior of the modeled object, all modeling conditions being the same; the model is convincingly presented for the SAH properties that are predicted using it.

From Fig. 6, it can be seen that the maximum air heating is observed in the interval between 13:00 and 14:00. This is due to the fact that the cosine of the angle between the sunray and the SAH surface in this time for the end of calendar autumn has a peak. A peak of $\sim 50^{\circ}\text{C}$ is attained by the air heating at a minimum inlet air velocity of 0.5 m/s. Results of the physical experiment for an inlet air velocity of 1.5 m/s are not shown in Fig. 6, since they have an absolute error of more than 28% and are not reliable, in the author's opinion.

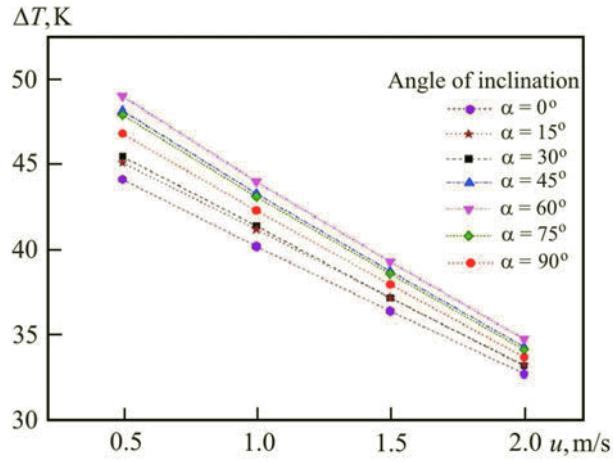


Fig. 5. Difference of the air temperatures at the SAH inlet and outlet vs. inlet velocity for different angles of inclination ($\Delta T = T_{\text{out}} - T_{\text{in}}$) of the SAH to the earth's surface.

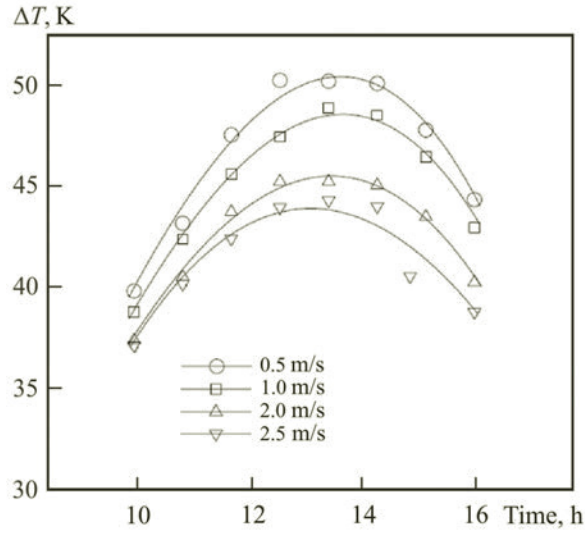


Fig. 6. Difference of the air temperatures at the SAH inlet and outlet during the day for different inlet air velocities.

Figure 7 gives the contours of air velocities and of static pressure in the solar heater. It can be seen from the figure that at the sites of location of the ribs, the air velocity increases 2–2.5 times compared to the region where there are no ribs. The heat-transfer coefficient increases as a result of the velocity increase, which in turn gives rise to the nonuniformity of the temperature contours [11, 12]. This fact is confirmed by the character of the temperature contours given in Fig. 4. However, the presence of the ribs exerts no influence, in practice, on the contour of static pressure in the SAH volume, which is also given in Fig. 7. At small values of the air velocity in the SAH volume (as small as 1–2 m/s), the presence of the ribs exerts no substantial influence on the aerodynamic characteristics of the flow.

Figure 8 shows the plot of behavior of the friction factor for the light-absorbing surface with ribs and without ribs. The friction factor in the present computer experiment was determined for the laminar regime of flow for all the velocities. The value of the friction factor is calculated from the formula

$$\lambda = \frac{\Delta P}{\frac{L}{d_{\text{eq}}} \rho \frac{u^2}{2}}. \quad (6)$$

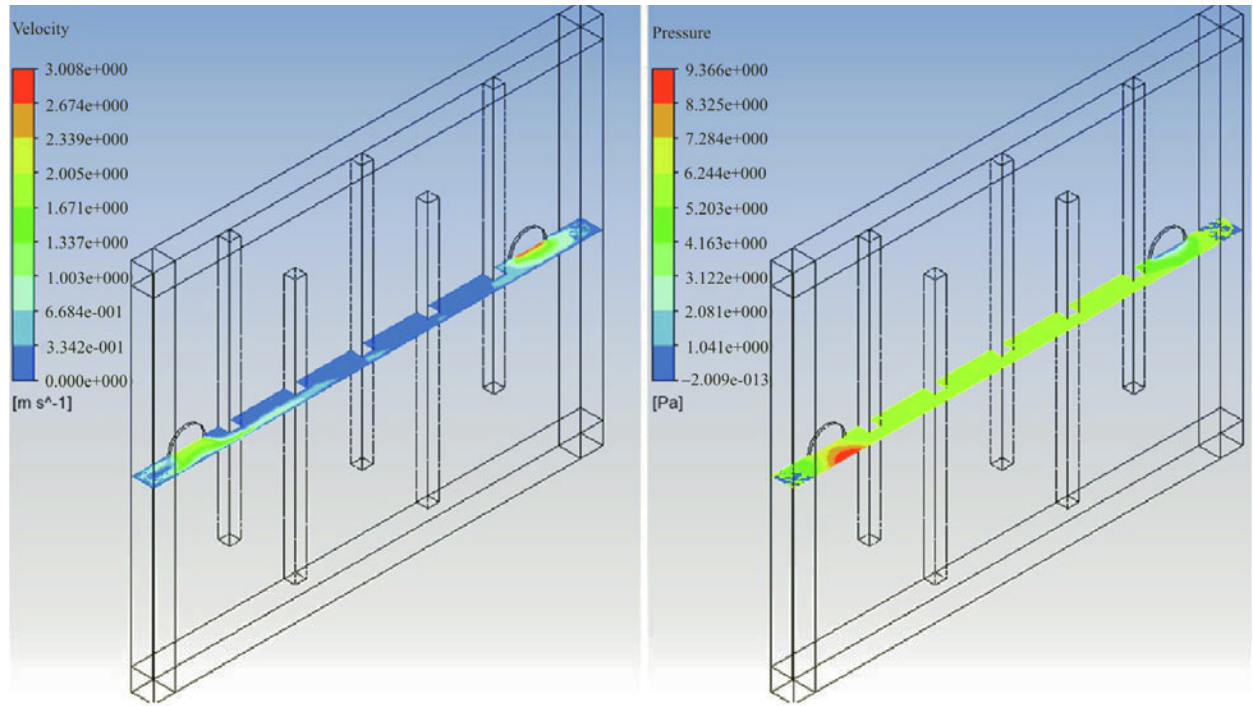


Fig. 7. Contours of air velocities and of static pressure in the SAH.

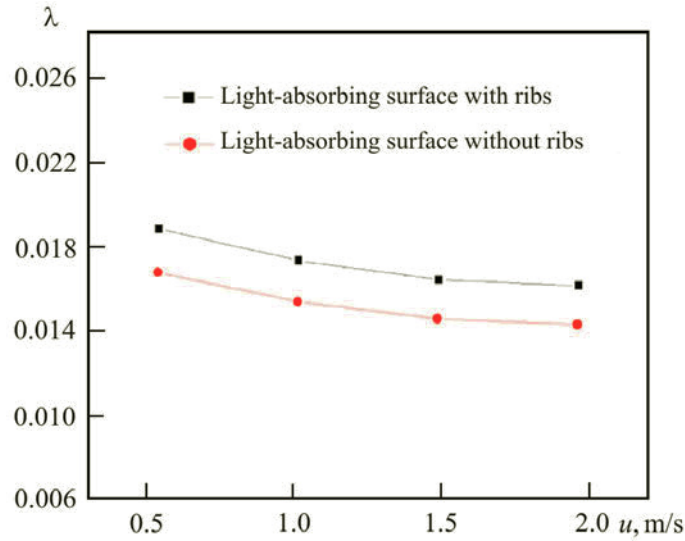


Fig. 8. Plot of behavior of the friction factor.

From Fig. 8 it can be seen that the presence of the ribs increases the friction factor, on the average, by 10–22% depending on the air velocity at entry into the SAH.

As has been noted above, among the advantages of an SAH are the low operating costs which are mainly ensured by the expenditure of energy in pumping the air. The presence of the ribs exerts no substantial influence on the operating costs, since the fan power is determined by the flow rate of the air and by its outlet pressure which is extremely low for the SAH in question. This fact follows from the determination of the fan power:

$$N = \frac{QP}{\eta} . \quad (7)$$

Conclusions. In the present study, we have constructed the computer model of a solar air heater in the ANSYS Fluent software product. The adequacy of the developed model has been confirmed by experimental data obtained in investigating the SAH in the city of Samara (53° north latitude and 50° east longitude) on the 22nd of November, 2016. On the basis of the resulting model, we conducted a computer experiment enabling us to establish the dependence of the difference of the air temperatures at the SAH inlet and outlet on the inlet air velocity for different angles of inclination ($\Delta T = T_{\text{out}} - T_{\text{in}}$) of the SAH to the earth's surface. According to the obtained data, the highest operating efficiency of the SAH is observed for the angle of inclination of the absorption surface to the earth $\alpha = 60^\circ$.

Also, we have obtained the dependence of the value of air heating throughout the daylight hours and have established that the maximum air heating is observed in the interval between 13:00 and 14:00. A peak of the air heating of $\sim 50^\circ\text{C}$ is attained at a minimum inlet air velocity of 0.5 m/s.

We have established the influence of the ribs on the heat exchange between the light-absorbing surface and the air. The curvilinear shape of the temperature contour is formed in the immediate vicinity of the ribs due to the formation of vortices and reverse-current zones. Investigation into the aerodynamic characteristics of the SAH has shown a minor influence of the ribs on the energy expenditure associated with the pumping of air in the air duct compared to the smooth light-absorbing surface.

NOTATION

a and a_t , molecular and turbulent thermal diffusivities respectively, m^2/s ; d_{eq} , equivalent diameter, m; L , characteristic length, m; P , assumed outlet pressure, Pa; Q , rate of flow of the air through the SAH, m^3/s ; T , temperature, K; u , air velocity, m/s; ΔP , pressure loss in the SAH, Pa; η , efficiency of the fan; λ , friction factor; μ , coefficient of dynamic viscosity, Pa·s; ρ , air density, kg/m^3 .

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